STATS 305C Applied Statistics III
Milestone 3: Latent Quality and Data-Mix Inference

Angikar Ghosal
Stanford University

Kaitlyn Wang
Stanford University

Goal and data. The feedback on Milestone 2 asked us to make the model actionable rather than only conclude that filters agree on some things and differ on others. We therefore keep the original hierarchical regression as a baseline, but add a latent-variable model whose output is a data-mix score. We use the same 120k-document modeling sample from the 2.26M-document WebOrganizer subset: 96,097 training documents, 23,903 held-out documents, 24 topics, 24 formats, and the top 500 domains plus an "other" bucket. The six complete scores are DCLM, rounded FineWeb-Edu, and PC ARC-Easy, PIQA, SciQ, and LAMBADA. Bounded scores are clipped at 10^{-4}, logit-transformed, and all scores/covariates are standardized on the training split. QuRater is excluded from the likelihood because it covers only 44 of the 100 sampled shards.

Generative model. Let $y_i \in \mathbb{R}^6$ be the transformed score vector and z_i contain an intercept, 17 text heuristics, topic indicators, format indicators, and domain indicators. We fit a two-factor latent filter model:

$$h_i \mid z_i, \Gamma \sim \mathcal{N}_2(\Gamma^\top z_i, I_2), \tag{1}$$

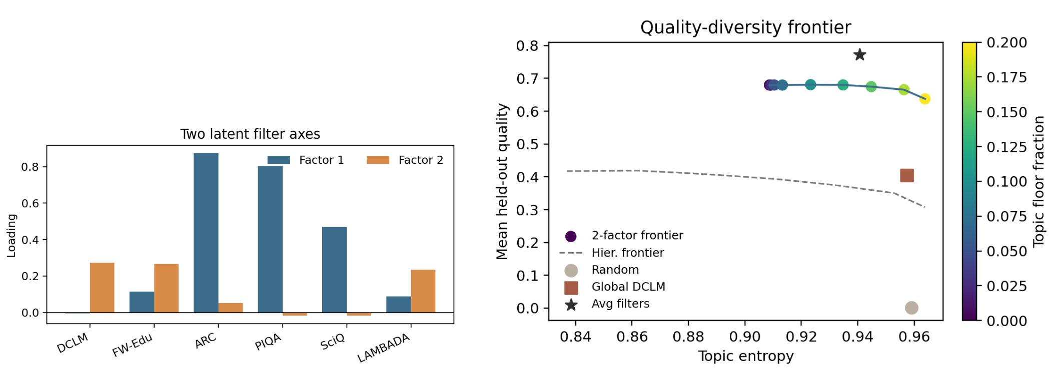
$$y_i \mid h_i, \alpha, \Lambda, \psi \sim \mathcal{N}_6(\alpha + \Lambda h_i, \operatorname{diag}(\psi)). \tag{2}$$

This says that filters are noisy measurements of two latent document axes whose prior means are predicted by observable document structure. The diagonal ψ keeps filter-specific noise explicit, while Λh_i captures shared cross-filter signal. We also fit a one-factor version; it mostly loaded on DCLM/FineWeb/LAMBADA and left the PC benchmark cluster unexplained, so the two-factor model is the scientifically useful specification.

Inference algorithm. We use EM, which is appropriate because the complete-data model is Gaussian. For fixed parameters, the E-step computes the posterior $p(h_i \mid y_i, z_i) = \mathcal{N}(m_i, V)$, where

$$V = (I + \Lambda^\top \operatorname{diag}(\psi)^{-1}\Lambda)^{-1}, \quad m_i = V\{\Gamma^\top z_i + \Lambda^\top \operatorname{diag}(\psi)^{-1}(y_i - \alpha)\}.$$

The M-step updates α, Λ, ψ by least squares using $\mathbb{E}[h_i]$ and $\mathbb{E}[h_i h_i^\top]$, and updates Γ by ridge regression of $\mathbb{E}[h_i]$ on z_i (equivalently a Gaussian prior on the covariate effects). The two-factor EM run converged in 180 iterations. The approximate observed-data log likelihood increased from $-746{,}210$ to $-678{,}610$, with final increment 0.067, so the optimization was stable rather than stopped manually. For held-out covariate-only prediction, RMSE is 0.898; conditioning on the observed filters gives reconstruction RMSE 0.625. We use the posterior expected average reconstructed score as the latent data-mix score.



(a) Estimated factor loadings. (b) Quality-diversity frontier.

Figure 1: The EM model discovers two interpretable filter axes and produces a data-mix frontier that dominates the previous hierarchical regression policy.

Posterior/factor analysis. Figure 1a shows the learned loadings. Factor 1 is a benchmark axis: ARC-Easy loads 0.87, PIQA 0.80, and SciQ 0.47. Factor 2 is a DCLM/FineWeb/LAMBADA axis, with loadings 0.27, 0.27, and 0.23 respectively. On the held-out set, Factor 1 correlates 0.71 with composite quality and Factor 2 correlates 0.45, so both axes matter for selection. The unique variances are smallest for ARC-Easy (0.064) and PIQA (0.186), meaning the benchmark axis is highly coherent, while DCLM, FineWeb-Edu, SciQ, and LAMBADA retain more filter-specific noise. Thus the model gives a sharper version of the Milestone 2 EDA: there is not a single universal quality filter, but there are two coherent latent axes. This also explains why the earlier single-response hierarchical regression improved prediction only slightly: averaging all filters collapses these two axes too early, while the factor model keeps both and recombines them only at the final decision stage.

Baseline comparison and actionability. We evaluate top-20% selection policies on held-out documents using the average of the six standardized scores as composite quality. Random selection has mean quality 0.001. Global DCLM reaches 0.405 with topic entropy 0.957. The previous hierarchical topic-floor policy reaches only 0.419, and its bootstrap difference from DCLM has a 95% interval crossing zero. The one-factor posterior policy improves to 0.448. The two-factor posterior policy reaches 0.680, and the two-factor topic-floor policy reaches 0.680 with all 24 topics covered and topic entropy 0.910. The simple average of all six filters is an upper-bound reference at 0.772; our model does not beat this tautological score, but it gives an interpretable probabilistic decomposition and a tunable data-mix frontier. We also separate two use cases: if only metadata and heuristics are available, the two-factor prior score is only 0.395 and does not improve on DCLM; if the six filters are available, the posterior score denoises and rebalances them. Thus the practical workflow is to score candidate documents with the available filters, infer posterior latent axes, and choose a mixture from the frontier rather than trusting a single global ranker.

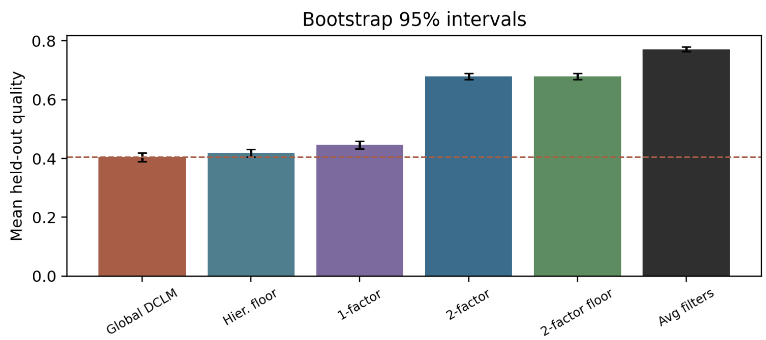


Figure 2: Bootstrap intervals over held-out rows. The old hierarchical policy is not clearly better than DCLM, but the two-factor policies are.

Quality-diversity result. Figure 1b varies the within-topic floor. At a 20% floor, the two-factor model reaches topic entropy 0.964, matching within-topic DCLM-level diversity, while still achieving quality 0.638, far above global DCLM's 0.405. The best frontier point is near a 10% topic floor: quality 0.681 and entropy 0.923. This is the recommended default if the goal is quality with some topic protection; the 20% floor is the safer choice if the pretraining mixture must look closer to the full corpus topic distribution. The bootstrap improvement of the two-factor topic-floor policy over DCLM is 0.274 standardized units, with 95% interval $(0.261, 0.291)$ (Figure 2). This directly answers the feedback: the model does something the EDA cannot. It estimates latent filter axes, then exposes a quality-diversity frontier for choosing a pretraining data mix.

Limitations. This still evaluates against held-out filter scores, not downstream LM loss. The next validation should train tiny language models on DCLM, average-filter, and two-factor-frontier mixtures at matched token budgets, then compare validation loss and benchmark accuracy. The posterior policy also uses the same six filters that define the composite target, so it should be interpreted as a principled smoothing and decomposition method, not proof that the composite is the right objective. Also, FineWeb-Edu is rounded ordinal and should eventually use an ordinal likelihood; QuRater should enter once all sampled shards are scored.

Reproducibility. The repository contains the full pipeline in numbered experiment scripts: factor EM fitting, policy evaluation, quality-diversity frontiers, bootstrap intervals, and figure generation. We fixed train/test splits and random seeds, stored generated artifacts separately from source code, and used the compiled figures in this report directly from those local runs. The experimental branch also contains staged commits for each modeling step, matching the milestone requirement that repository history document progress.

AI assistance. We used Codex and Claude Code for code generation, debugging, and report drafting. Numerical claims were checked against locally generated artifacts.